Quizzes

约 1431 个字 8 张图片 预计阅读时间 5 分钟

Quiz 1

1. Represent the decimal number 75 as a 7-bit binary number: `( 1 )`, and as a BCD code: `( 2 )`. Then add an even parity bit at the most significant bit (MSB) of the BCD code to form a 9-bit code: `( 3 )`.
2. Simplify the Boolean expression $XY + \bar{X}Z + X\bar{Y} + \bar{Y}Z$ to its minimum number of literals:
   - A: $Y + Z$
   - B: $X + Z$
   - C: $X + Y$
   - D: $X + Y + Z$
3. Which of the following expressions has its dual equal to its complement?
   - A: $\bar{A} + BC$
   - B: $\bar{A}B + BC$
   - C: $\bar{A}\bar{B} + BC$
   - D: $\bar{A}B + A\bar{B}$
4. What is the sum-of-minterms (SOM) form of the expression $F(A, B, C) = \bar{A}B + BC$?
   - A: $\sum m(2, 3, 7)$
   - B: $\sum m(0, 1, 4)$
   - C: $\sum m(2, 4, 6)$
   - D: $\sum m(2, 3, 6)$

Answer

1. As below:
   - **Binary Code**: 1001011_2
     - $75 = 64 + 11$
   - **BCD Code**: $0111\ 0101$
   - **9-bit Code (Even Parity)**: $1\ 0111\ 0101$
2. B ($X + Z$)
3. D ($\bar{A}B + A\bar{B}$)
   - $f^D(x, y, z) = \bar{f}(x, y, z) = f^D(\bar{x}, \bar{y}, \bar{z}) \iff f(x, y, z) = f(\bar{x}, \bar{y}, \bar{z})$
4. A ($\sum m(2, 3, 7)$)

Quiz 2

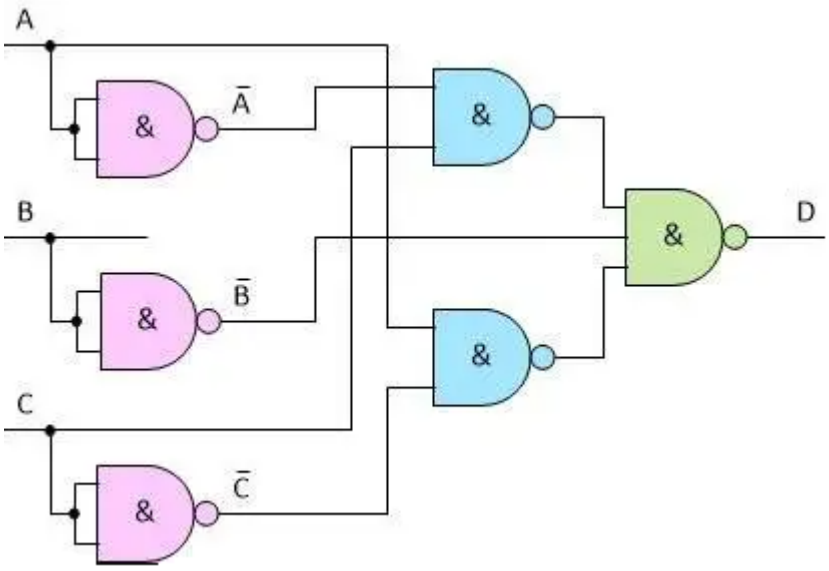
1. For the function $F = AB(C + D) + C(BD' + \bar{A}D)$, the **literal cost** is: `( 1 )`, and the **gate input cost with NOT (GN)** is: `( 2 )`.
2. Find the essential prime implicants for $F(W, X, Y, Z) = \sum m(0, 1, 4, 6, 7, 8, 9, 12, 14, 15)$:
   - A: $Y'Z', XZ'$
   - B: $\bar{X}\bar{Y}, XY$
   - C: XY, XZ'
   - D: $Y'Z', \bar{X}\bar{Y}$
3. Find a minimal sum-of-products (SOP) expression for the function F with the don't-care conditions: $F(A, B, C, D) = \sum m(1, 5, 6, 13, 14) + \sum d(4, 12)$:
   - A: $B\bar{C} + \bar{A}\bar{C}D + BCD'$
   - B: $B\bar{D} + B\bar{C} + \bar{A}\bar{C}$
   - C: $\bar{C}D + B\bar{C} + B\bar{D}$
   - D: $B\bar{C} + B\bar{D} + \bar{A}\bar{C}D$
4. How many implicants, prime implicants (PI), and essential prime implicants (EPI) are there in $F(A, B, C) = \sum m(0, 4, 5) + \sum d(1)$?
   - A: 8, 1, 1
   - B: 7, 1, 1
   - C: 6, 1, 1
   - D: 5, 3, 1

Answer

1. As below:
   - **Literal cost**: 9
   - **Gate input cost with NOT (GN)**: 17
2. B ($\bar{X}\bar{Y}, XY$)
3. D ($B\bar{C} + B\bar{D} + \bar{A}\bar{C}D$)
4. A (8, 1, 1)
   - **Implicants**: 3 + 4 + 1 = 8.
   - **PI/EPI**: Combined term $\bar{C}$ covers all required minterms.

Quiz 3

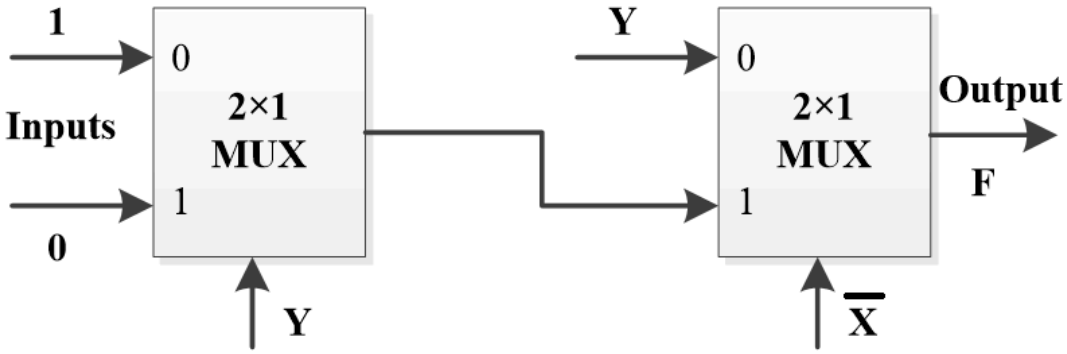
1. Find the t_{PHL} from input C to the output D, assuming $t_{\text{PHL}} = 0.20\ \text{ns}$ and $t_{\text{PLH}} = 0.36\ \text{ns}$ for each gate.



   - A: 0.56 ns
   - B: 0.6 ns
   - C: 0.76 ns
   - D: 0.92 ns
2. Consider a 4-input priority encoder with inputs D_3, D_2, D_1, D_0. The encoder responds to the most significant 1 (highest priority to D_3). It produces outputs A_1, A_0 and a valid flag V, where $V = 1$ indicates at least one input is 1. What is the logic expression for V?
   - A: $D_3D_2 + D_1D_0$

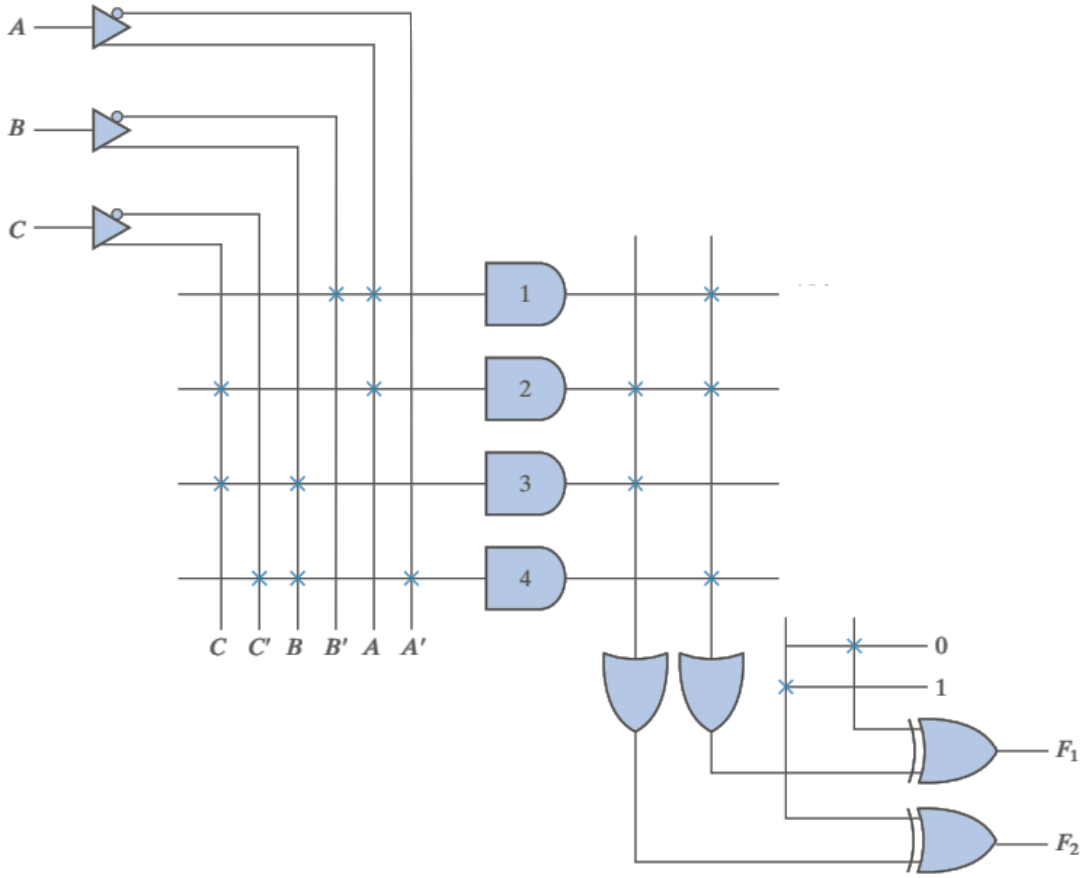
- B: $D_3D_2 + D_1 + D_0$
- C: $D_3D_2D_1 + D_0$
- D: $D_3 + D_2 + D_1 + D_0$

3. Please find the logic expression of output F in the figure.



- A: $XY + X\bar{Y}$
- B: $\bar{X}\bar{Y} + XY$
- C: $\bar{X}Y + XY$
- D: $\bar{X}Y + X\bar{Y}$

4. What specific Boolean function does the PLA implement for F_2?



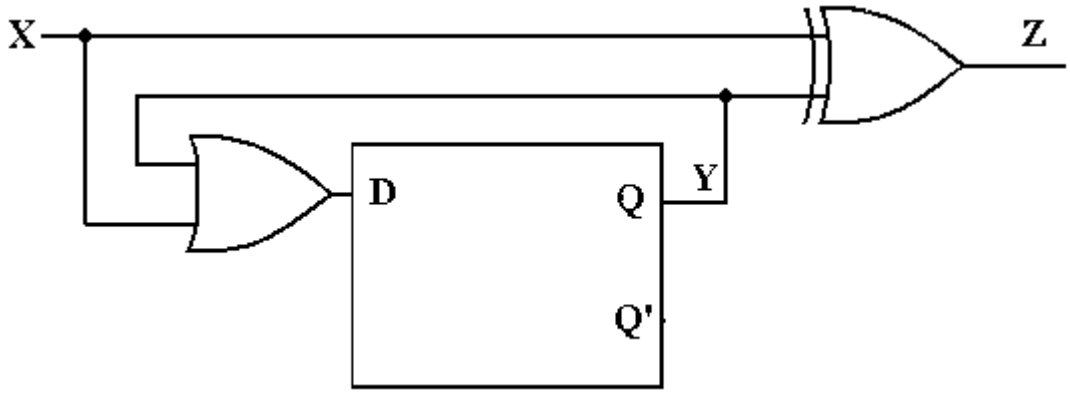
- A: $AC + BC$
- B: $\bar{C} + \bar{A}\bar{B}$
- C: $\bar{A}C + BC$
- D: $AC + B$

Answer

1. C (0.76 ns)
2. D ($D_3 + D_2 + D_1 + D_0$)
3. B ($\bar{X}\bar{Y} + XY$)
4. B ($\bar{C} + \bar{A}\bar{B}$)

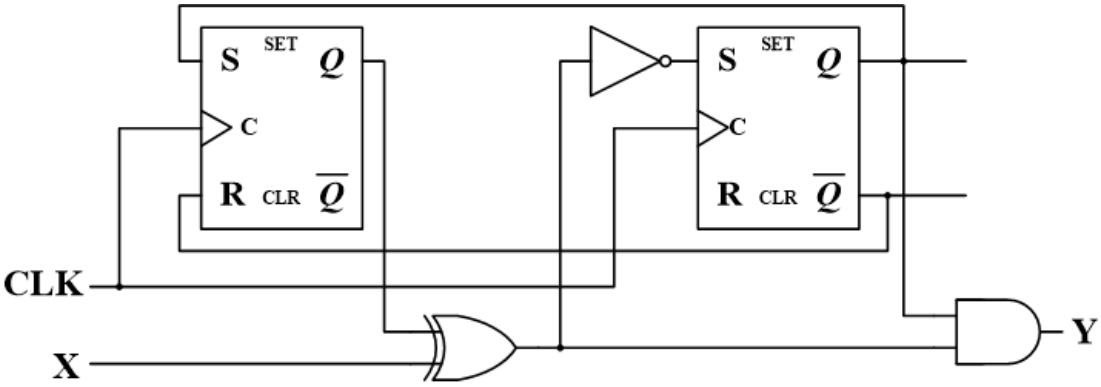
Quiz 4

1. For the decimal number $(-32)_{10}$ represented as an 8-bit signed binary number:
   a. The signed-magnitude representation is `( 1 )`;
   b. The signed-1's complement representation is `( 2 )`;
   c. The signed-2's complement representation is `( 3 )`.
2. Given two 4-bit signed 2's complement numbers, $A = (0100)_2$ and $B = (1101)_2$:
   a. The result of $A - B$ is `( 1 )`;
   b. Does overflow occur?
3. Which of the following statements about latch and flip-flop behavior is **CORRECT**?
   - A: Metastable state in a latch occurs when both inputs (S and R) are held at logic 0, causing the outputs to oscillate indefinitely.
   - B: Oscillation in a basic SR latch happens when both inputs are 0 and then return to 1 simultaneously, leading to an undefined or endless switching state if gate delays are equal.
   - C: 1s catching is a problem in edge-triggered D flip-flops, where a brief glitch on the D input during the clock edge can cause an incorrect output.
   - D: A metastable state is a stable, long-term condition that can be intentionally used to store an intermediate logic value.
4. As the sequential circuit shown below, determine its input equation, output equation, and next state equation.



- A: $D = X \oplus Y, Z = X + Y, Q(t+1) = D$
- B: $D = X + Y, Z = X \oplus Y, Q = D(t+1)$
- C: $D = X + Y, Z = X \oplus Y, Q(t+1) = D$
- D: $D = X \oplus Y, Z = X + Y, Q = D(t+1)$

5. The timing parameters for the gates and flip-flops are as follows: NOT Gate: $t_{pd} = 0.75$ ns; XOR Gate: $t_{pd} = 3.0$ ns; AND Gate: $t_{pd} = 1.5$ ns; Flip-flop: $t_{pd} = 3.0$ ns, $t_s = 1.25$ ns, $t_h = 0.5$ ns. Find the longest path delay from positive clock edge to an external output, and the maximum frequency of operation of the circuit.



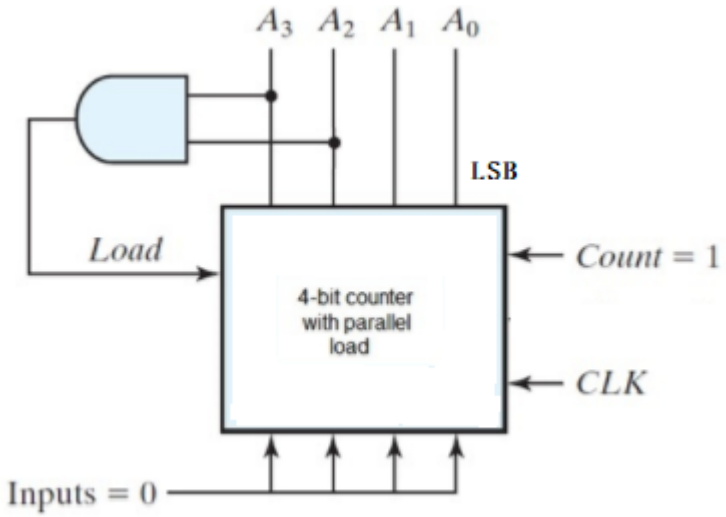
- A: 7.75 ns, 130 MHz
- B: 7.5 ns, 120 MHz
- C: 4.75 ns, 180 MHz
- D: 7.5 ns, 125 MHz

Answer

1. 10100000, 11011111, 11100000
   - $32 = (00100000)_2$.
   - Signed-magnitude: sign bit 1 plus magnitude 0100000 gives 10100000.
   - 1's complement: invert 00100000 gives 11011111.
   - 2's complement: add 1 to 11011111 gives 11100000.
2. 0111, overflow does not occur
   - $A = 4, B = -3$, so $A - B = 4 - (-3) = 7 = (0111)_2$.
3. B
4. C ($D = X + Y, Z = X \oplus Y, Q(t+1) = D$)
5. D (7.5 ns, 125 MHz)
   - Clock-to-output longest path: $3.0 + 3.0 + 1.5 = 7.5$ ns.
   - Register-to-register critical period: $3.0 + 3.0 + 0.75 + 1.25 = 8.0$ ns, so $f_{\max} = 125$ MHz.

Quiz 5

1. What is the function of the following 4-bit counter with parallel load?



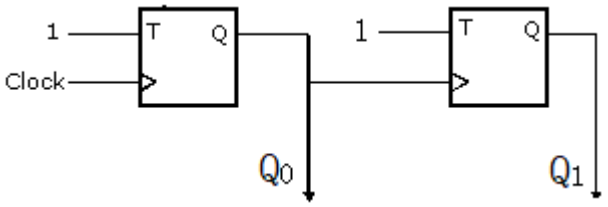
   - A: It counts from 0 to 12 and back to 0.
   - B: It counts from 0 to 9 and back to 0.
   - C: It counts from 0 to 10 and back to 0.
2. Given two 8-bit registers $R1$ and $R2$, with $R1 = (1011\ 0110)_2$ and $R2 = (0111\ 1010)_2$. After two microoperations:

a. $R1 \leftarrow sl\ R1$;

b. $R1 \leftarrow R1 \oplus R2$.

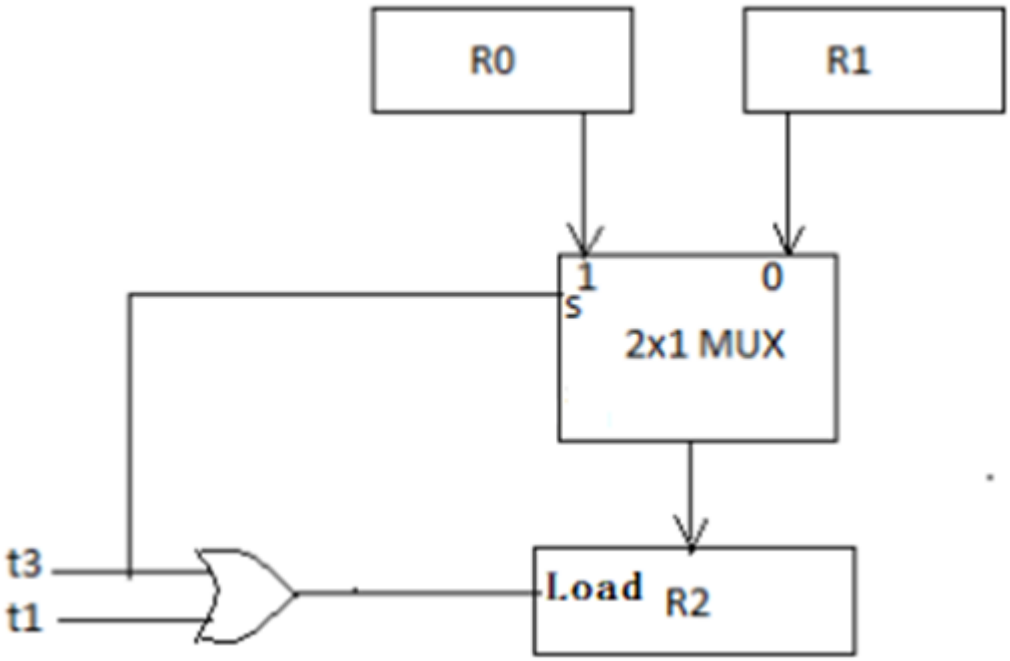
Then $R1 =$ `( ? )` . - A: $0001\ 0110$ - B: $0011\ 0110$ - C: $1101\ 1110$ - D: $0001\ 1110$

3. In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00, what are the next four values of Q_1Q_0?



- A: 11, 10, 01, 00
- B: 10, 11, 01, 00
- C: 10, 00, 01, 11
- D: 11, 10, 00, 01

4. Choose the correct register transfer operations for the following diagram.



- A: $t_1 : R2 \leftarrow R1;\ t_3 : R2 \leftarrow R0$
- B: $t_1 : R2 \leftarrow R1;\ t_1't_3 : R2 \leftarrow R0$
- C: $t_1t_3' : R2 \leftarrow R1;\ t_3 : R2 \leftarrow R0$

5. You are designing a synchronous sequential circuit to detect the sequence "101" on a serial input stream. What is the minimum number of states required for a Mealy-type circuit and a Moore-type circuit, respectively?

- A: Mealy: 3 states; Moore: 3 states

- B: Mealy: 4 states; Moore: 4 states
- C: Mealy: 3 states; Moore: 4 states
- D: Mealy: 4 states; Moore: 5 states

Answer

1. A
   - The load signal is asserted when $A_3A_2 = 11$, so the counter loads 0000 after reaching $1100_2 = 12$.
2. A ($0001\ 0110$)
   - Shift left: $1011\ 0110 \rightarrow 0110\ 1100$.
   - XOR with $R2$: $0110\ 1100 \oplus 0111\ 1010 = 0001\ 0110$.
3. A (11, 10, 01, 00)
   - Both T flip-flops have $T = 1$, so they toggle. Starting from $Q_1Q_0 = 00$, the ripple sequence is $11 \rightarrow 10 \rightarrow 01 \rightarrow 00$.
4. C
   - t_1 enables load and selects input 0 of the MUX, so $R2 \leftarrow R1$ when t_1t_3'.
   - t_3 enables load and selects input 1 of the MUX, so $R2 \leftarrow R0$.
5. C
   - A Mealy detector can assert output on the transition that completes "101", requiring 3 states.
   - A Moore detector needs an extra output state after the sequence is detected, requiring 4 states.

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评论区